

Network Flow III – Some Problems

(ADA 2023, CSE 222)

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Outline

- ① Max-Flow Min-Cut Theorem and Bipartite Matching
- ② Proof of Hall's Theorem
- ③ Assigning jobs to people

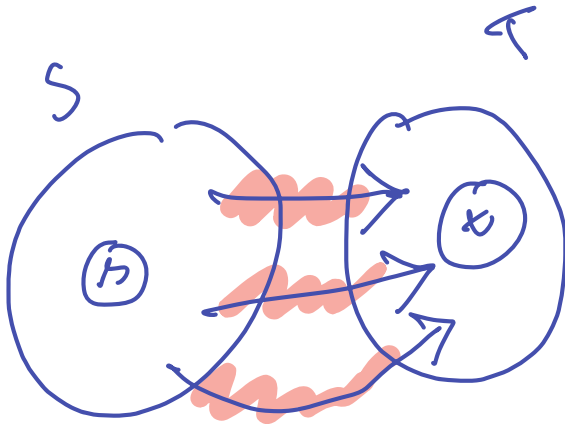
Max-Flow Min-Cut Theorem

Let (G, s, t, c) be a flow network and $f: E(G) \rightarrow \mathbb{Z}^+$ be a valid (s, t) -flow. Then, the followings are equivalent.

- 1 $\text{val}(f)$ is a maximized.
- 2 G_f has no augmenting paths.
- 3 There exists an (s, t) -cut (S, T) such that $c(S, T) = \text{val}(f)$.

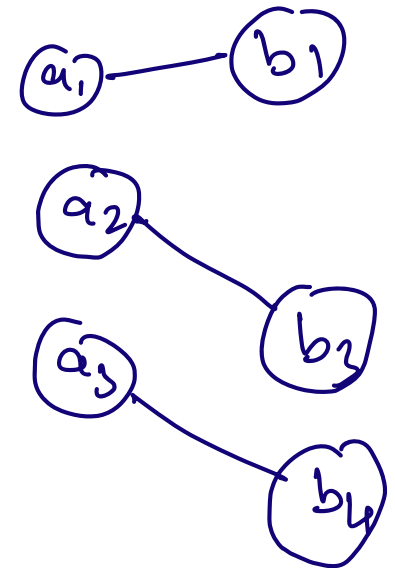
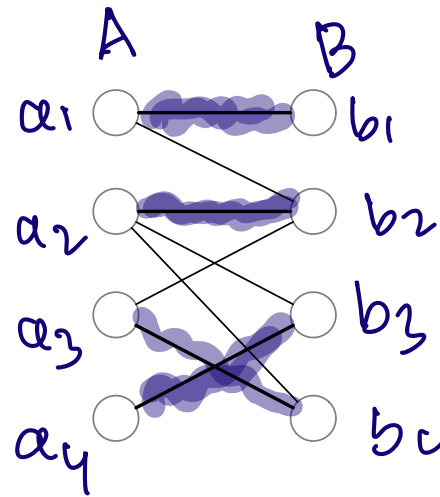
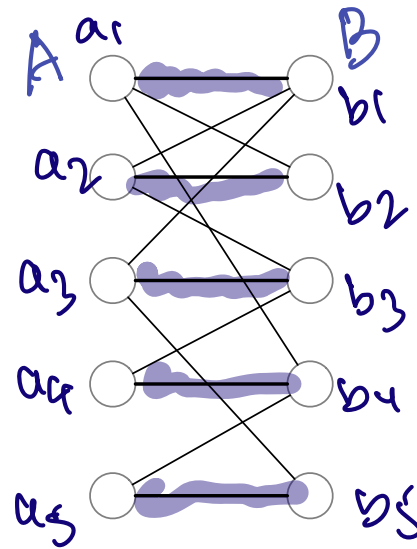
$$\text{val}(f) = \sum_{u: (s, u) \in E(G)} f(s, u)$$

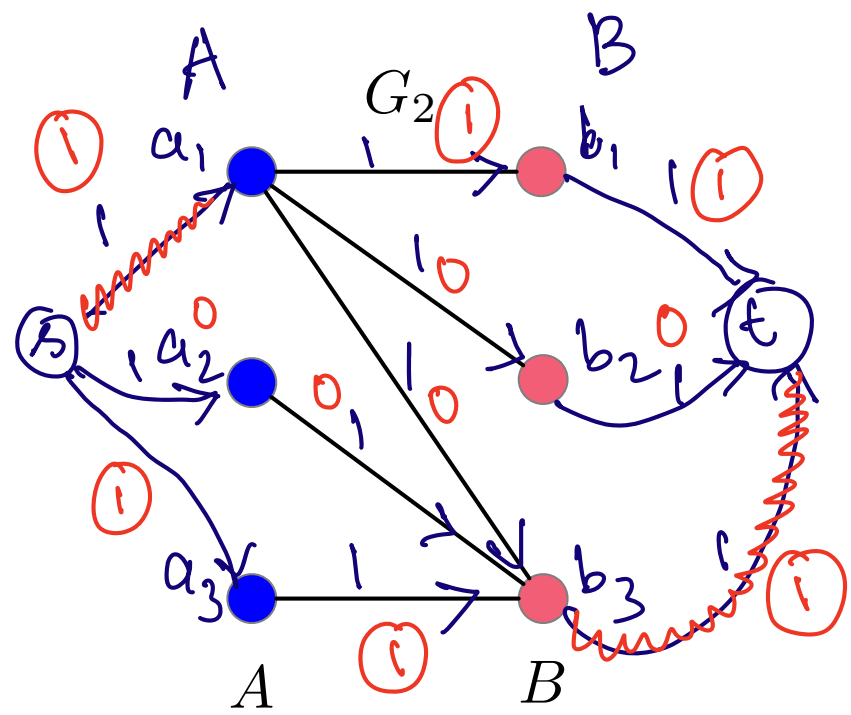
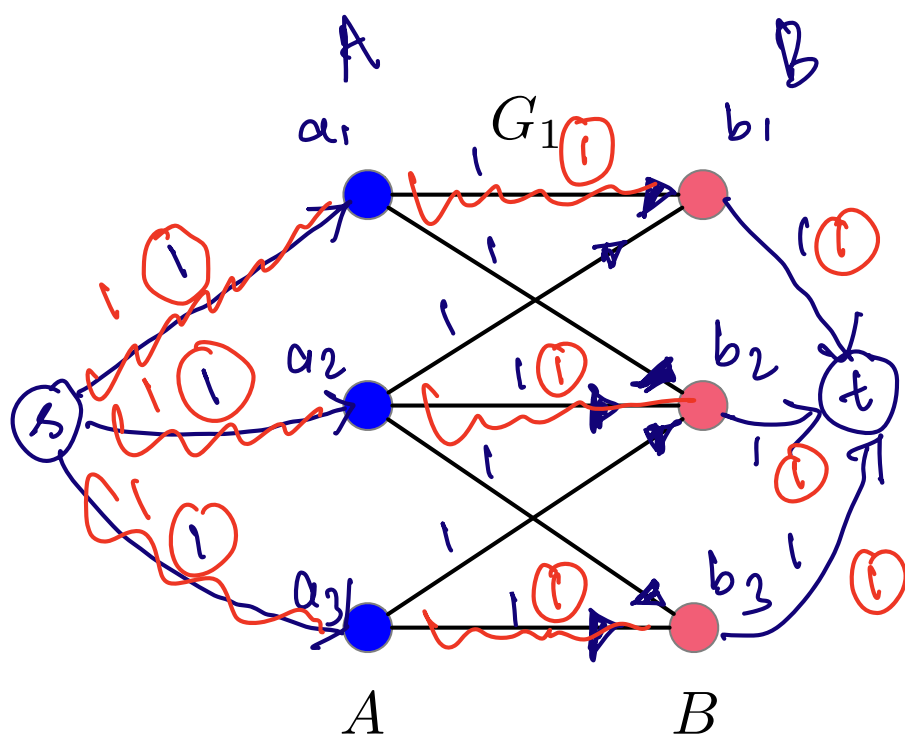
This particular (s, t) -cut (S, T) is an (s, t) -cut with minimum possible capacity.

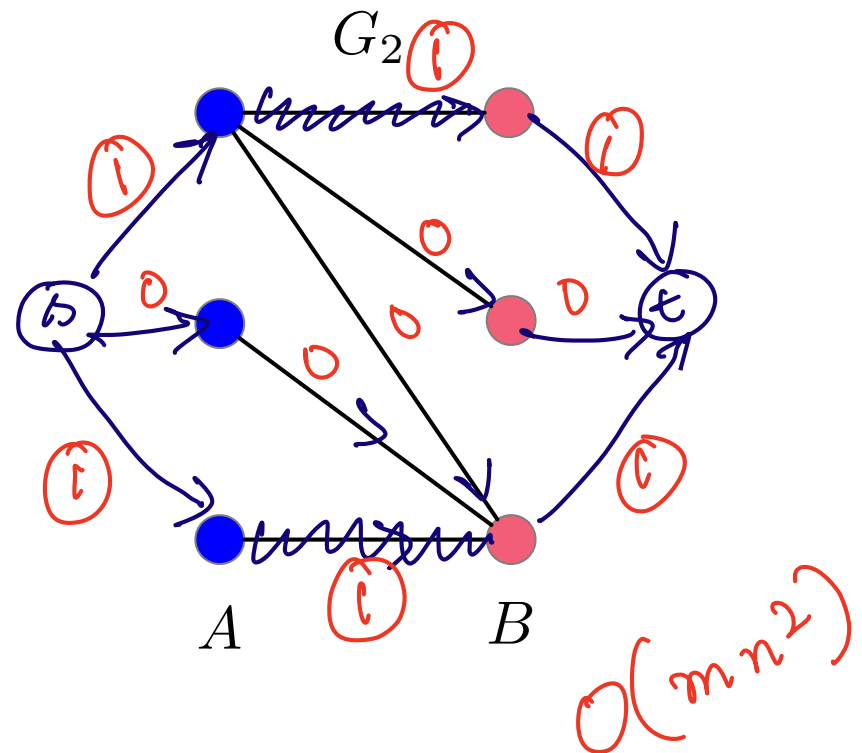
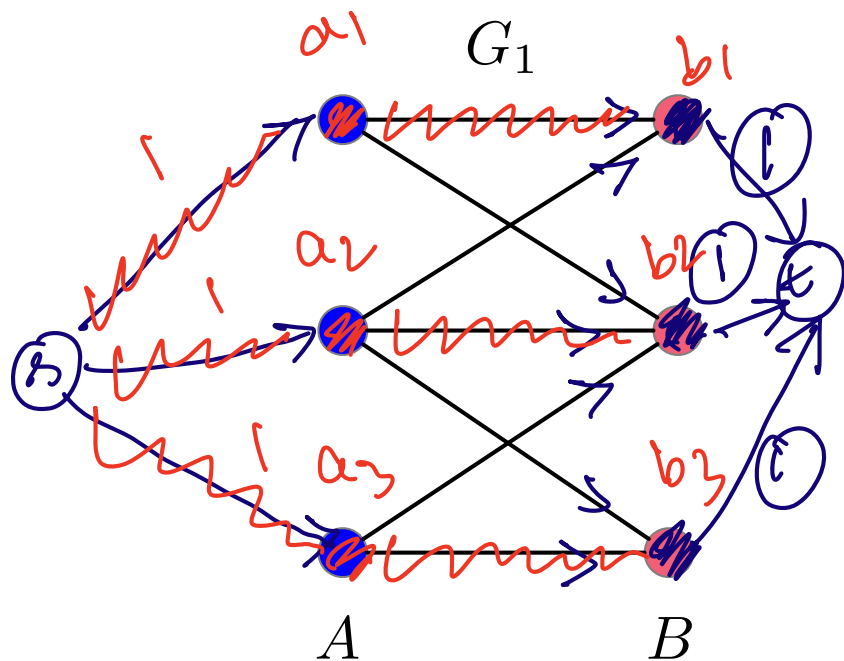


Bipartite Matching Problem

- **Input:** A bipartite graph $G = ((A, B), E)$.
- **Goal:** A set of edges $M \subseteq E(G) \subseteq A \times B$ such that no two edges share a common endpoint and $|M|$ is maximized.







- Let k be the value of a maximum s - t flow in this network.
- M' has k edges.
- At most one edge of M' is an outgoing edge from every vertex of A .
- At most one edge of M' is an incoming edge into every vertex of B .

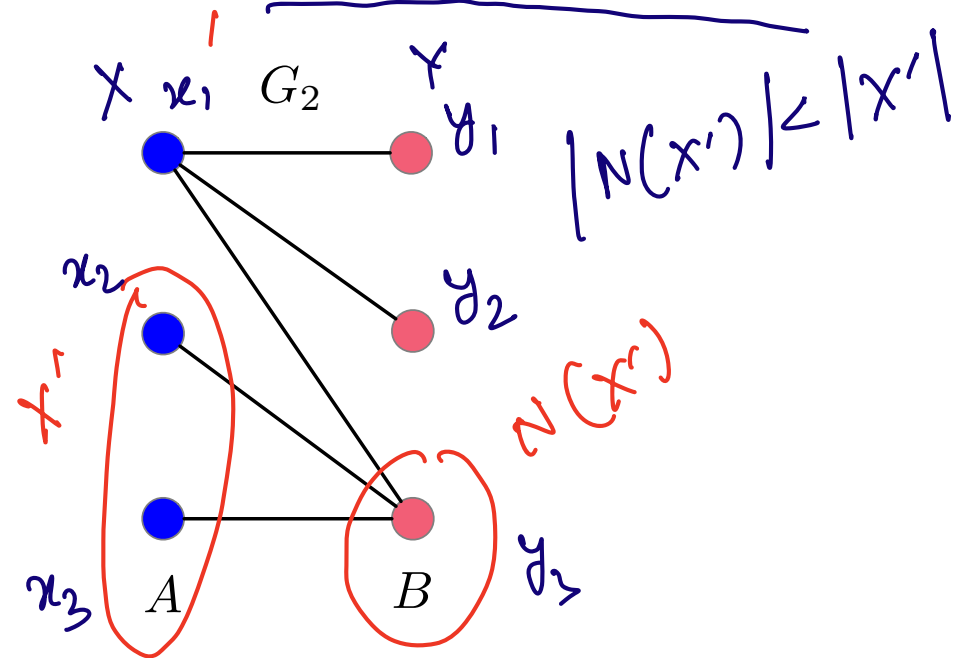
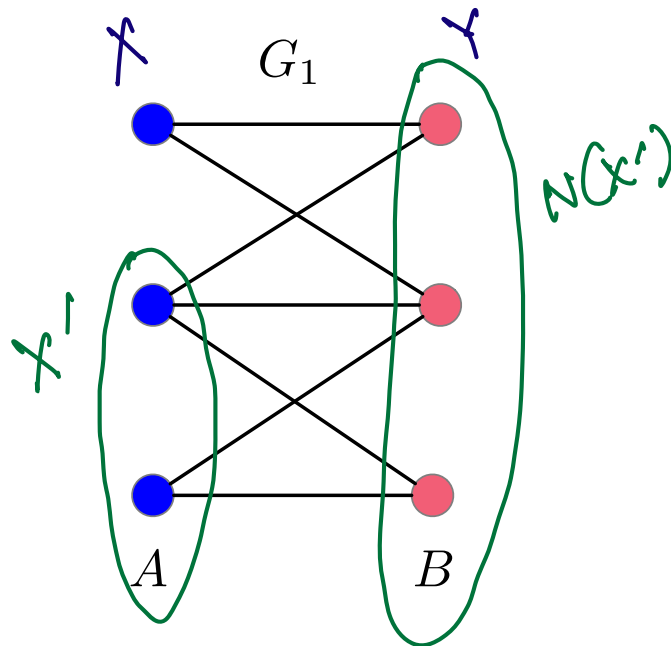
$O(mn^2)$

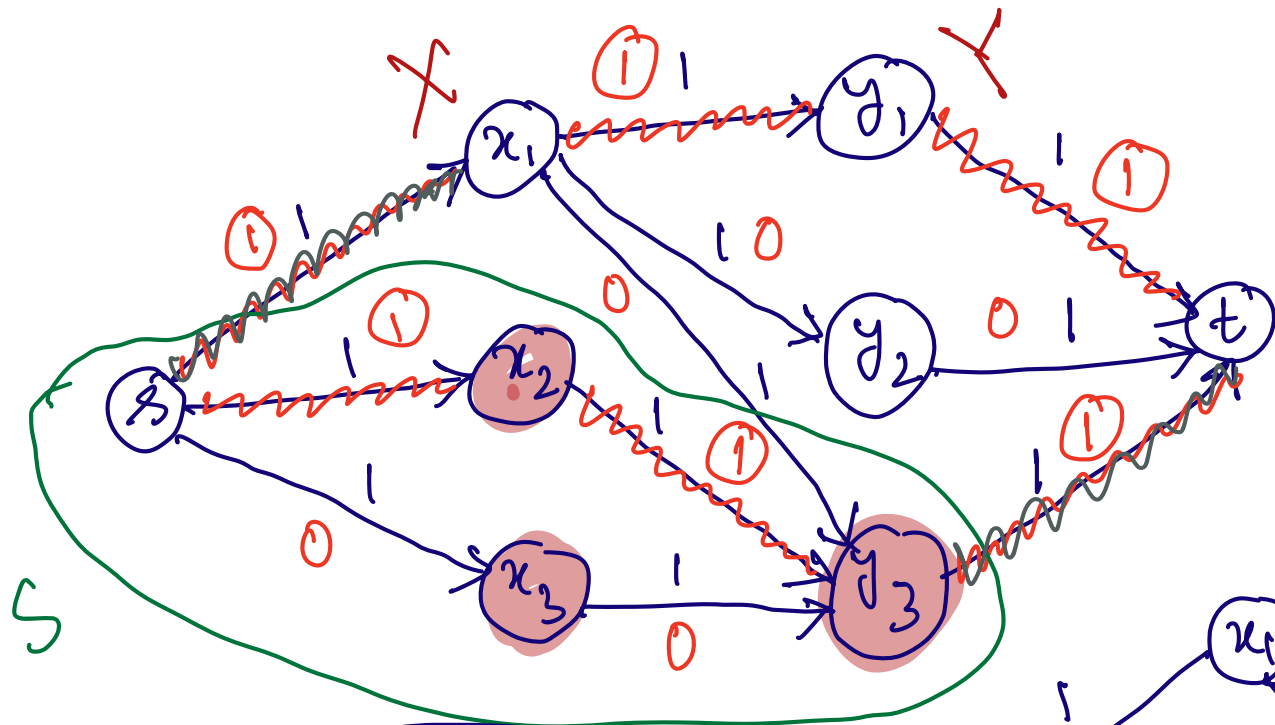
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Hall's Theorem

- Let $G = ((X, Y), E)$ be a graph with. Then a matching M in G is called a *perfect matching* if all vertices of G participates in a M .
- Hall's Theorem:** Let $G = ((X, Y), E)$ be a bipartite graph with $|X| = |Y|$. Then, G has a perfect matching if and only if for all $X' \subseteq X$, $|N(X')| \geq |X'|$.



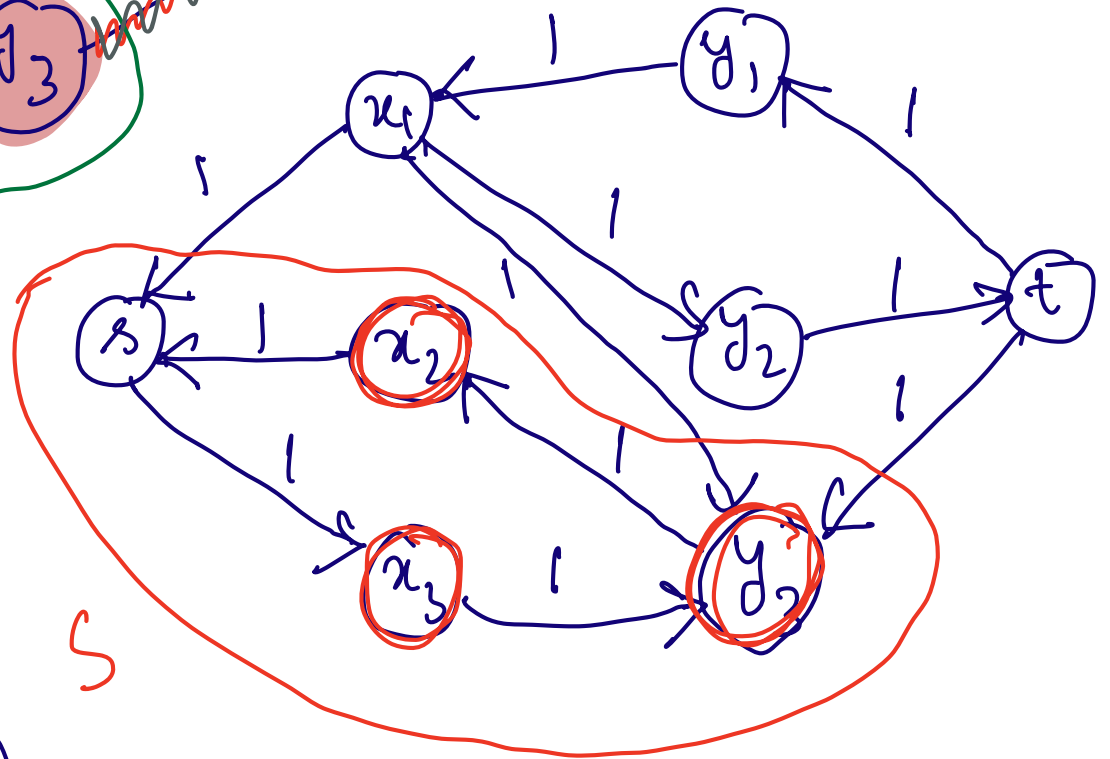


$$x' = \{x_2, x_3\}$$

$$N(x') = \{y_3\}$$

$$S = \{s, x_2, x_3, y_3\}$$

$$T = \{x_1, y_1, y_2, t\}$$



$$c(S, T) = c(s, x_1) + c(y_3, t)$$

$$f(S, T) = f(s, x_1) + f(y_3, t) - f(x_1, y_3)$$

$$S = \{x\} \cup x' \cup N(x')$$

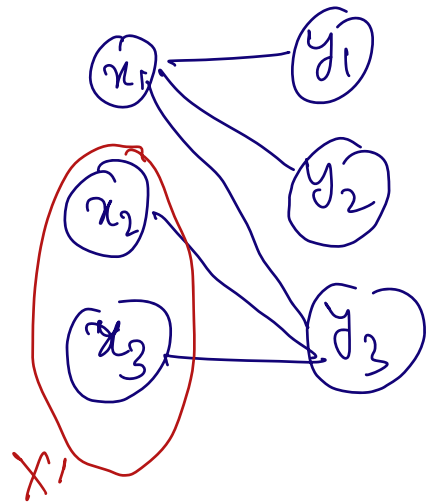
$$T = V(G) \setminus S.$$

$$S \cap X = x'$$

$$S \cap Y = N(x')$$

$$T \cap X = X \setminus x'$$

$$T \cap Y = Y \setminus N(x')$$

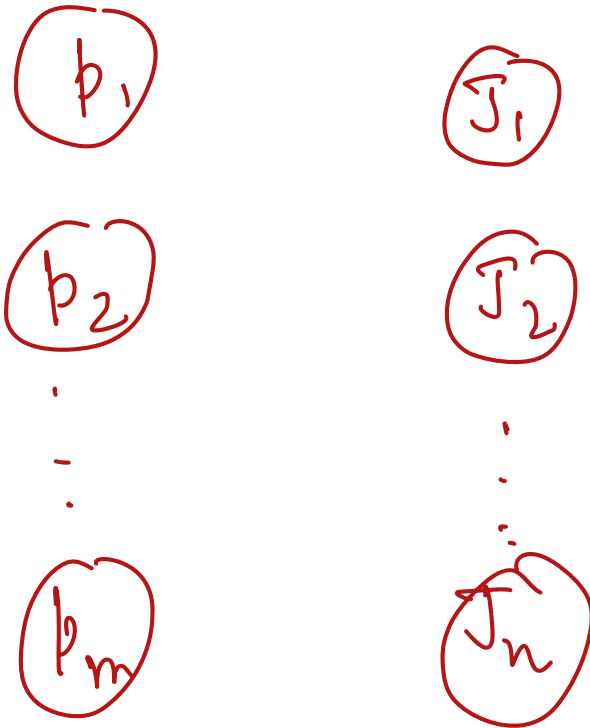


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Assigning job to people

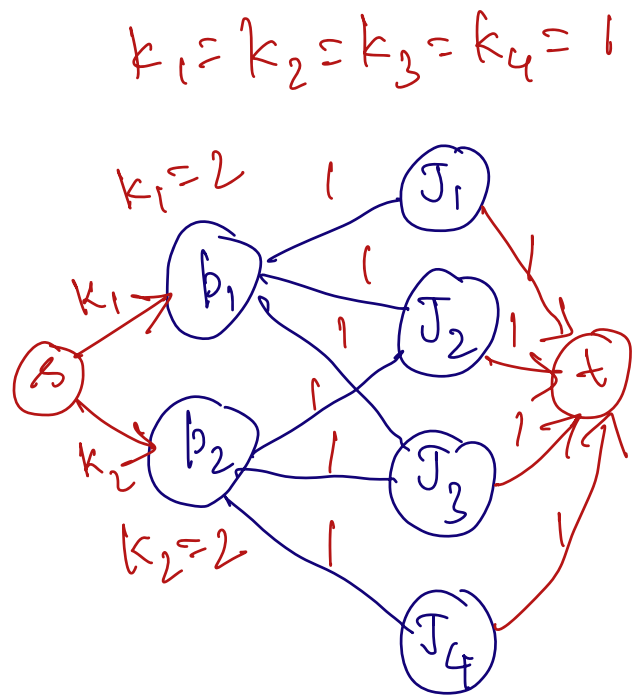
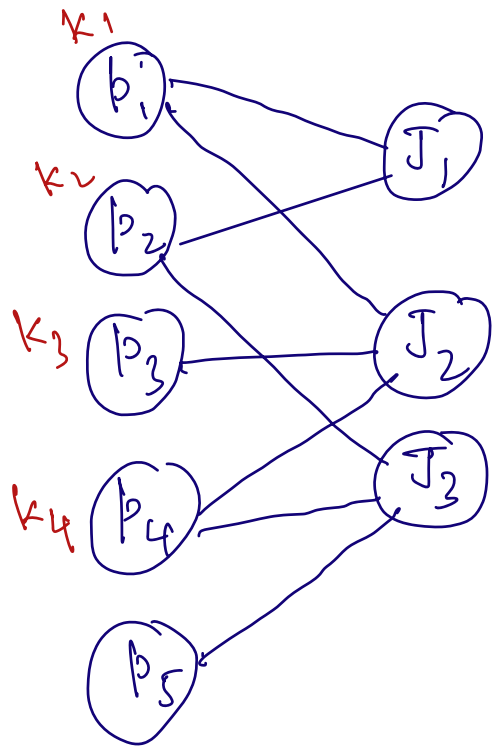
- **Input:** There are n jobs and m people (persons). For each job, there is a set of people who can do that job. For each person j , there a limit on the number of jobs is k_j .
- **Goal:** Find an assignment of jobs to people so that every job is assigned to some people, and no person is overloaded.



person j is
assigned at most
 k_j jobs.

If p_i can do job
 J_k , then

(p_i, J_k) is
an edge.



Maximum flow value = #jobs

if and only if every job can be
assigned to some people.

Assigning job to people

- **Input:** There are n jobs and m people (persons). For each job, there is a set of people who can do that job. For each person j , there a limit on the number of jobs is k_j .
- **Goal:** Find an assignment of jobs to people so that every job is assigned to some people, and no person is overloaded.
- **Construction of DAG:** $G = X \cup Y \cup \{s, t\}$ where X is the set of persons, and Y is the set of jobs.
- Add directed edges (s, i) for all $i \in X$. Capacities of all these edges are 1. $(s, p_i) \quad p_i \in X$
- If job i can be done by the set of persons $Y_i \subseteq Y$, then add directed edges (i, j) for all $j \in Y_i$. Capacities of all these edges are 1. $X_i \subseteq X$
- For all $j \in Y$, add edges (j, t) with capacity k_j . (p_j, T_i)
- **Theorem:** There is an assignment if and only if there is an (s, t) -flow of value n in G .

(Necessary and sufficient condition)

Some additional announcements

- Quiz-4 (MCQ based) on 17th April and Regular Lecture with problem solving on 13th April.
- **Some specific decisions based on mid-sem feedback:**
 - As discussed already in a previous lecture, after mid-sem exam, we will continue to problem solving part in every lecture.
 - **Relaxed evaluation in Quiz Component:** Best 1 out of 2 pre-mid sem quizzes and best 1 out of 2 post midsem quizzes.
 - **Only one problem in Theory Assignment-3:** The assignment will be floated in the week of 17th April. We will give only “one problem from network flow” and will give 5 days to submit.
 - Last two lectures will specifically dedicate to tutorial style problem solving. Those two lectures would give some insights on the flavor of questions in the end-sem exam.